

Due: August 17, 2026

1. Consider the following linear system:

$$\begin{aligned}2x + w &= 7 \\3x + 2y + 3z &= -2 \\2x + 3y - 4z &= 3 \\x + 3z &= 5\end{aligned}$$

- Find the coefficient matrix.
 - Write the linear system in matrix form.
 - Find the augmented matrix.
2. Write the following linear system as a linear combination of the columns of the coefficient matrix:

$$\begin{aligned}2x - 3y + 5z &= -2 \\x + 4y - z &= 3\end{aligned}$$

3. For the following matrices, determine whether it is in reduced row echelon form (RREF), row echelon form (REF), or neither:

a)

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

c)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

4. Solve the linear system with the following augmented matrix using elementary row operations

(Gauss-Jordan reduction):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 3 \\ 0 & 1 & 1 & 1 \end{array} \right]$$

5. Check whether the matrix $A = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ is singular or nonsingular. If it is nonsingular, find its inverse A^{-1} .

6. For the following matrices, find the inverses if they exist:

a)

$$\begin{bmatrix} 1 & 3 \\ -2 & 6 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

7. Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$.

a) Find A^{-1} .

b) Find $(A^T)^{-1}$ and verify that $(A^T)^{-1} = (A^{-1})^T$.

8. Evaluate the determinants of the following matrices:

a)

$$\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$$

b)

$$\begin{bmatrix} 0 & 3 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & -5 \end{bmatrix}$$

c)

$$\begin{bmatrix} 4 & 2 & 2 & 0 \\ 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

9. Verify that $\det(AB) = \det(A) \cdot \det(B)$ for:

$$A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 3 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 5 \\ 2 & 1 & 3 \end{bmatrix}$$

10. Compute the inverse of the following matrix using the adjoint method ($A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$):

$$A = \begin{bmatrix} 1 & 2 & -3 \\ -4 & -5 & 2 \\ -1 & 1 & -7 \end{bmatrix}$$

11. Find all eigenvalues λ for which:

$$\det \left(\begin{bmatrix} \lambda - 1 & -4 \\ 0 & \lambda - 4 \end{bmatrix} \right) = 0$$